

WHAT IS CRISPR-Cas?

DA-CBC's mission is to utilize the capabilities of CRISPR-Cas technology to make a difference in our farmers' lives and ensure our land's sustainability. We explore how CRISPR-Cas evolves from a scientific discovery to becoming a tool for innovation. Our aim is to increase crop yield, enhance crop resilience, and create a more food-secure tomorrow.

Join us as we cultivate a greener and more prosperous Philippines- **ONE CROP AT A TIME.**

CRISPR-Cas or **Clustered Regularly Interspaced Short Palindromic Repeats-CRISPR-associated protein** helps researchers modify a plant's genome in a very precise manner. This can make targeted changes to the plant's DNA such as deleting or inserting specific DNA sequences or genes resulting in crops with the intended traits.

CRISPR-Cas Crops Underway



Wheat that does not contain gluten (Sánchez-León et al.)



Tomatoes with increased levels of vitamin C and resistance to bacterial spot disease (Lemmon et al., Zsögönet et al.)



Rice with increased yield, improved quality, nutrition, and pest and disease resistance (Miao et al.)



Eggplants that are non-browning (Sagarbarria et al.)



Citrus fruits with resistance to citrus greening disease (Nian Wang)



Cacao with disease resistance (Fister et al.)



CRISPR-Cas IN CROP BIOTECHNOLOGY

CRISPR-Cas enables scientists to enhance desirable traits in crops, such as the following:

- Crop resilience to extreme environments: Improve tolerance to extreme and stressful conditions such as drought
- Pest and disease resistance: Protect crops from devastating pests and diseases without the use of harmful chemicals
- Better nutrition: Develop crops with higher nutritional content; addressing malnutrition.
- Higher yield: Boost crop productivity to meet growing demands.



IMPROVING LIVES ONE CROP AT A TIME

DA-CBC

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CRISPR-Cas

Revolutionizing Agriculture

